

Joining $\rightarrow$ Employee { Empname, Emp-no., salary, age, Dept-no. }
Department { Dept-no., D-name, D-loc, No. of emp }

Q $\rightarrow$ Find the name of the employee and their salary and working city.

Π E.Empname, E.salary, D.Dloc

(Employee (E)  Department (D))
E.Deptno. = D.Deptno.

Select E.Empname, E.salary, D.Dloc from Employee E X Department D
Where E.Deptno. = D.Deptno.;

Outer Join! → Left outer

Right outer

Both way outer.



R1

A	B	C
a ₁	b ₁	c ₁
a ₂	b ₂	c ₂
a ₃	b ₃	c ₃

R2

C	D	E
c ₄	d ₁	e ₁
c ₂	d ₂	e ₂
c ₃	d ₂	e ₃

Left Outer! → R1 ~~R2~~

A	B	C	D	E
a ₁	b ₁	c ₁	NULL	NULL
a ₂	b ₂	c ₂	d ₂	e ₂
a ₃	b ₃	c ₃	d ₂	e ₃

Right outer Join! → R1 ~~R2~~

A	B	C	D	E
NULL	NULL	c ₄	d ₁	e ₁
a ₂	b ₂	c ₂	d ₂	e ₂
a ₃	b ₃	c ₃	d ₂	e ₃

Both way outer Join! →

R1 ~~R2~~

A	B	C	D	E
a ₁	b ₁	c ₁	NULL	e ₂
a ₂	b ₂	c ₂	d ₂	e ₃
a ₃	b ₃	c ₃	d ₂	e ₁
NULL	NULL	c ₄	d ₁	e ₁

θ-Join!

~~θ~~
θ.

R1

a	d
5	7
9	12
14	13

R2

b	c
7	10
15	11
8	17

R1 ~~θ~~ R2
a < b

a	d	b	c
5	7	7	10
5	7	15	11
5	7	8	17
9	12	15	11
14	13	15	11

A	B	C	D	E
NULL	NULL	c ₁	d ₁	e ₁
a ₁	b ₂	c ₂	d ₂	e ₂
a ₃	b ₃	c ₃	d ₂	e ₃
a ₁	b ₁	c ₁	NULL	NULL

Divide Operation $\rightarrow R1 \div R2$

R

a	b	c	d
a	b	c	d
b	c	e	b
c	d	c	d
c	d	e	b
a	a	d	e

S

c	d
e	b

$R \div S$

$\left. \begin{matrix} a \\ c \end{matrix} \right\} \begin{matrix} b \\ d \end{matrix}$

a	b
c	d

a	b
c	d

a	b
b	d
c	d

$\begin{matrix} a & b \\ b & c \end{matrix} \quad \begin{pmatrix} c & d \\ e & b \end{pmatrix}$

$\left. \begin{matrix} a \\ a \\ c \\ c \end{matrix} \right\} \begin{matrix} b \\ b \\ d \\ d \end{matrix} \right\}$

TRC (Tuple Relational Calculus)

$\{t_1.A_1, t_2.A_2, \dots, t_n.A_n \mid \text{COND}(t_1, t_2, t_3, \dots, t_{n+1}, t_{n+2}, \dots, t_{nm})\}$

$\exists \quad \forall$

Q. List the name and address of the employee who are works in a 'Research' Department?

$\{ \underline{t.name}, \underline{t.add} \mid \underline{\text{Employee}(t)} \text{ AND } (\underline{\exists d})$
 $(\underline{\text{Department}(d)} \text{ AND } \underline{d.dname = 'Research'}$
 $\text{AND } \underline{d.dnum = t.Dno.}) \}$

Q.2.1 List the name and Designation of those employees whose salary is more than 10000.

→ { t.name, t.Design | Employee(t) AND t.sal > 10000 }

Q.3. Find the name of the employee and his/her manager name using TRC.

{ t.name, m.name | Employee(t) AND Employee(m) AND
t.Emprno. = m.mgrno. }



→ 001 SSK A.P. 120000 005
002 - - -
003 - - -
004 - - -
005 SPN AP 15800 100.

